Lecture 3
Binary Images and Morphological Image Processing

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Binary image processing

- Binary images are common
  - Intermediate abstraction in a gray-scale/color image analysis system
    - Thresholding/segmentation
    - Presence/absence of some image property
  - Text and line graphics, document image processing
- Representation of individual pixels as 0 or 1, convention:
  - foreground, object = 1 (white)
  - background = 0 (black)
- Processing by logical functions is fast and simple
- Morphological image processing has been generalized to gray-level images via level sets

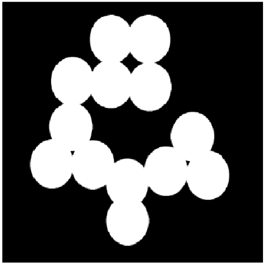
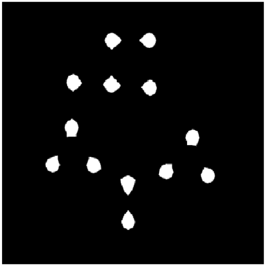
like classification problem

more care about the shape!

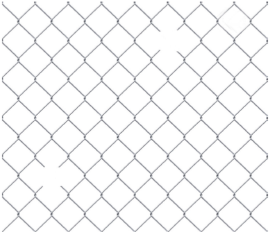
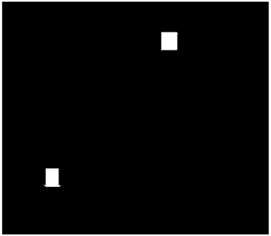
1. It is a tool for extracting image components that are useful in the representation and description of region shape, such as boundaries, skeletons, etc.
2. We use the language in set theory to represent morphological operations.

Examples

- Blob separation

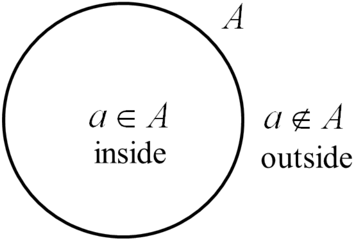



- Fence hole detection

Basic Concepts from Set Theory I

1. Let A be a set in Z^2 ($2D$ integer space)
2. If $a = (a_1, a_2)$ is an element of A, where a_1 and a_2 are coordinates of pixels in $2D$, then we write $a \in A$.
3. If a is not an element of A, then we write $a \notin A$.



4. If A is a set with no element, then we call it *null or empty set*,

$$A = \emptyset.$$

Basic Concepts from Set Theory II

5. A set is specified by the contents of two braces: $\{\}$,
   - a. e.g. $C = \{w | w = -d, \text{for } d \in D\}$.
   - b. C represents a set of elements.
   - c. w represents the elements.
   - d. All elements inside C are formed by multiplying the element coordinates, d, by -1.
6. A is a subset of B: every element of a set A is also an element of another set B. (every element of A is in B),

$$A \subseteq B.$$

7. Union of two sets A and B: the set of all elements belonging to either A or B. (all elements are either in A or B)

$$C = A \cup B.$$

Basic Concepts from Set Theory III

8. Intersection of two sets A and B: the set of all elements belonging to both A and B. (all elements are in A and B)

$$D = A \cap B.$$

9. Two sets A and B are disjoint or mutually exclusive if their intersection is an empty set,

$$A \cap B = \emptyset.$$

10. Complement of a set A: the set of elements not contained in A,

$$A^c = \{w | w \notin A\}.$$

11. Difference of two sets A and B: the set of all elements belonging to A and not contained in B,

$$A - B = \{w | w \in A, w \notin B\} = A \cap B^c.$$

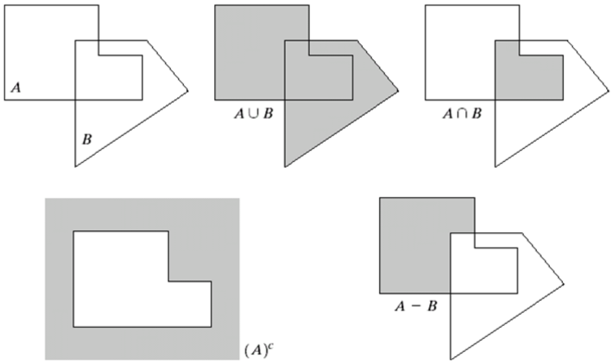


FIGURE 9.1 (a) Two sets A and B. (b) The union of A and B. (c) The intersection of A and B. (d) The complement of A. (e) The difference between A and B.

Basic Concepts from Set Theory V

12. Reflection of set B:

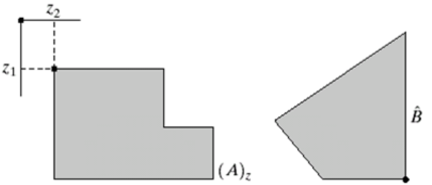
上下左右反轉

$$\hat{B} = \{w | w = -b, \text{for } b \in B\}.$$

13. Translation of set A:

平移

$$(A)_z = \{c | c = a + z, \text{for } a \in A\}.$$



a b

FIGURE 9.2 (a) Translation of A by z. (b) Reflection of B. The sets A and B are from Fig. 9.1.

Logic Operations Involving Binary Images

- All images are binary images.
- In images, black indicates a binary 1 and white indicates a 0.
- Operations are limited to binary variables.

0,1 不用緊, consistant is fine

TABLE 9.1
The three basic logical operations.

p	q	p AND q (also $p \cdot q$)	p OR q (also $p + q$)	NOT (p) (also $\bar{p}$)
0	0	0	0	1
0	1	0	1	1
1	0	0	1	0
1	1	1	1	0

Logic Operations Involving Binary Images: cont.

- The principal logic operations used in image processing are: AND, OR, NOT (COMPLEMENT).
- These operations are functionally complete.
- Logic operations are preformed on a pixel by pixel basis between corresponding pixels (bitwise).
- Other important logic operations : XOR (exclusive OR), NAND (NOT-AND)
- Logic operations are just a private case for a binary set operations, such : AND Intersection , OR Union, NOT-Complement

Examples

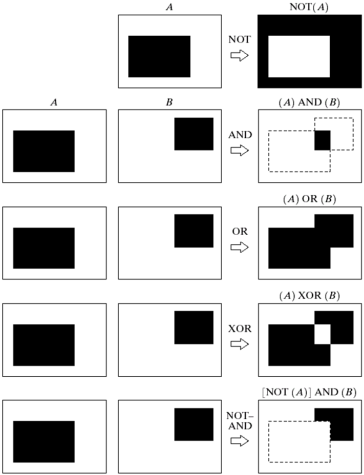


FIGURE 9.3 Some logic operations between binary images. Black represents binary 1s and white binary 0s in this example.

Black = 1
White = 0

shrink the original image

1. Let A and B be sets in Z^2.
2. The erosion of A by B is defined as

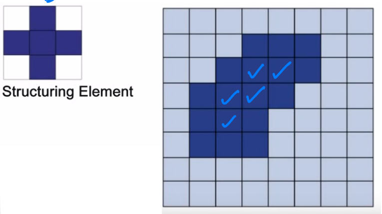
$$A \ominus B = \{z | (B)_z \subseteq A\}.$$

All pixels must be in original image!!!

any translation!

3. The set of all displacements, z, such that B, translated by z, is contained in A.
4. An example

B



"Erode" the skin

cannot tell the computer "tear" off the skin!

review: https://www.youtube.com/watch?v=fmyE7DiaIYQ

Erosion II

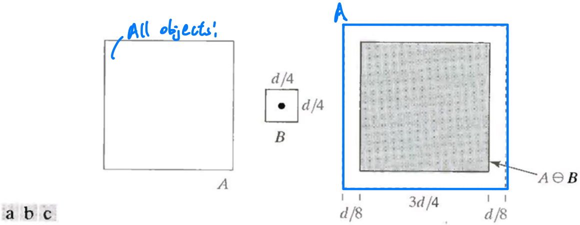


FIGURE 9.6 (a) Set A. (b) Square structuring element. (c) Erosion of A by B, shown shaded

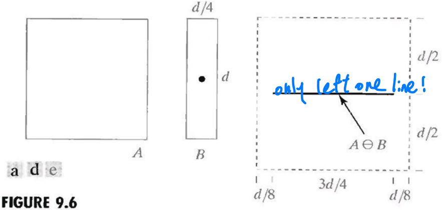


FIGURE 9.6
(a) Set A.(d) Elongated structuring element. (e) Erosion of A using this element.

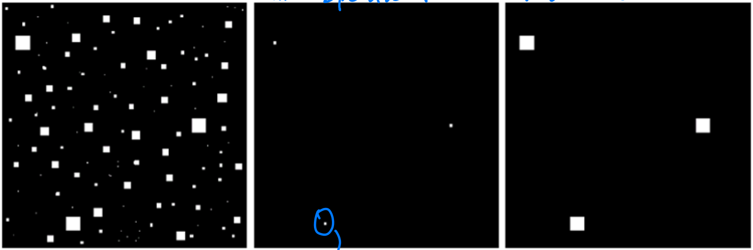
Erosion III

5. Dashed lines show the original set for reference.
6. Solid lines/shared region (in Figs c and e) show the limit beyond which any further displacements of the origin of B by z would cause the erosion set not contained in A.
7. z is in $A \ominus B$ when B is completely contained by the set A.

Erosion: application

1. Eliminating irrelevant detail from a binary image.
2. In the images, black indicates a binary 0 and white indicates a 1.
3. All elements in the structuring element have the same binary values as the objects of interest. feature selection!

like specific faces   recover it!



only take centre   also can convolution

a b c

FIGURE 9.7 (a) Image of squares of size 1, 3, 5, 7, 9, and 15 pixels on the side. (b) Erosion of (a) with a square structuring element of 1's, 13 pixels on the side. (c) Dilation of (b) with the same structuring element.

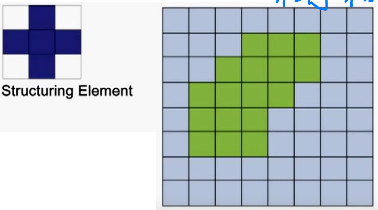
Dilation

1. Let A and B be sets in Z^2.
2. The dilation of A by B is defined as

$$A \oplus B = \left\{ z | (\hat{B})_z \cap A \neq \emptyset \right\}.$$

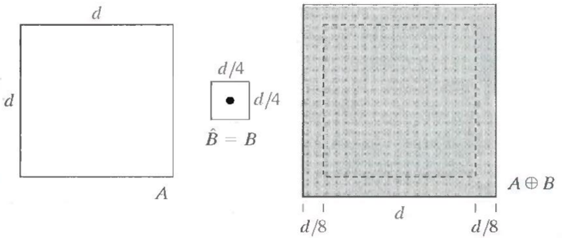
3. The set of all displacements, z, such that $\hat{B}$ and A overlap by at least one element.
4. Similar to concept of convolution mask. Why?
5. An example

任何一格也可!



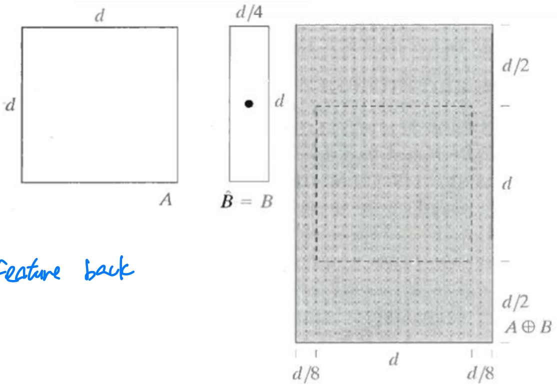
review: https://www.youtube.com/watch?v=x03ED27rMHs a b c

FIGURE 9.4
(a) Set A.
(b) Square structuring element (dot is the center).
(c) Dilation of A by B, shown shaded.



a d e

(d) Elongated structuring element.
(e) Dilation of A using this element.



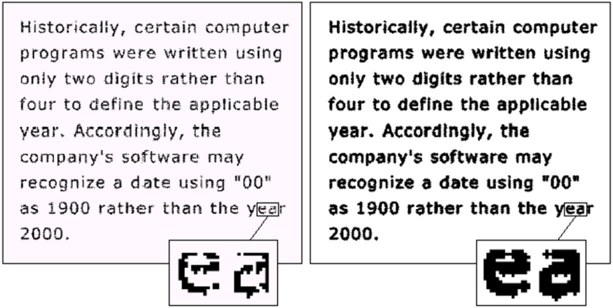
label ⇒ put feature back

Notes

1. Dashed lines (in Figs. c and e) show the original set for reference.
2. Solid lines (in Figs c and e) show the limit beyond which any further displacements of the origin of $\hat{B}$ by z would cause the intersection of $\hat{B}$ and A to be empty.
3. z is in $A \oplus B$ when A and $\hat{B}$ overlap by at least one element.

Dilation: application

Bridging gaps in broken characters:



a c
b

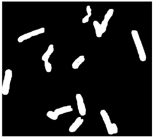
FIGURE 9.5
(a) Sample text of poor resolution with broken characters (magnified view).
(b) Structuring element.
(c) Dilation of (a) by (b). Broken segments were joined.

$$B = \begin{array}{|c|c|c|} \hline 0 & 1 & 0 \\ \hline 1 & 1 & 1 \\ \hline 0 & 1 & 0 \\ \hline \end{array}$$

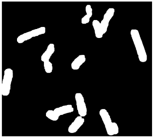
Comparision between dilation and erosion

- dilation
  - Expands the size of 1-valued objects
  - Smoothes object boundaries
  - Closes holes and gaps

Original (701x781)



dilation with
3x3 structuring element



dilation with
7x7 structuring element

- erosion
  - Shrinks the size of 1-valued objects
  - Smoothes object boundaries
  - Removes peninsulas, fingers, and small objects

Original (701x781)

erosion with
3x3 structuring element

erosion with
7x7 structuring element

Dual operations

- Dilation and erosion are duals of each other with respect to set complementation and reflection.

$$(A \ominus B)^c = A^c \oplus \hat{B}.$$

- Dilation expands objects (represented by '1') in an image and erosion shrinks objects.

Opening and Closing

- Opening smoothes contours , eliminates protrusions
- Closing smoothes sections of contours, fuses narrow breaks and long thin gulfs, eliminates small holes and fills gaps in contours
- These operations are dual to each other
- These operations are can be applied few times, but has effect only once

Opening I

1. Recall: Dilation expands an object and erosion shrinks it.
2. In general, opening smoothes the object contour.

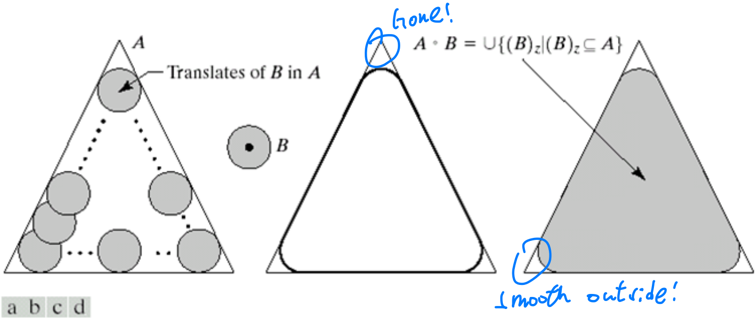


FIGURE 9.8 (a) Structuring element B "rolling" along the inner boundary of A (the dot indicates the origin of B). (c) The heavy line is the outer boundary of the opening. (d) Complete opening (shaded).

Opening II

3. Definition

$$A \circ B = (A \ominus B) \oplus B.$$

4. It represents erosion of A by B, followed by dilation of the result by B.
5. It means that it takes the union of all translates of B that fit into A. (Imagine you roll the ball inside the object)
6. Subimage property: $A \circ B$ is a subset (subimage) of A.
7. Convergence property

$$(A \circ B) \circ B = A \circ B.$$

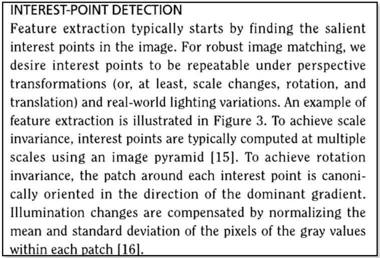
8. Opening tends to remove some of the forground pixels from the edges, but less destructive than erosion.

remove!

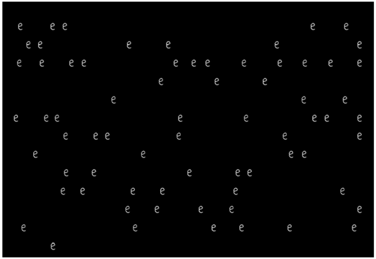
example of opening: recognition

Binary image f

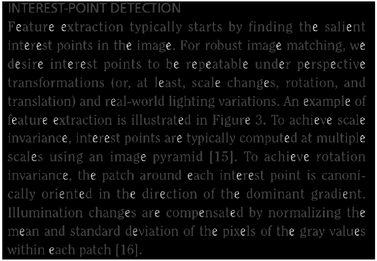
1400


INTEREST-POINT DETECTION
Feature extraction typically starts by finding the salient interest points in the image. For robust image matching, we desire interest points to be repeatable under perspective transformations (or, at least, scale changes, rotation, and translation) and real-world lighting variations. An example of feature extraction is illustrated in Figure 3. To achieve scale invariance, interest points are typically computed at multiple scales using an image pyramid [15]. To achieve rotation invariance, the patch around each interest point is canonically oriented in the direction of the dominant gradient. Illumination changes are compensated by normalizing the mean and standard deviation of the pixels of the gray values within each patch [16].

2000



same size
no noise!!!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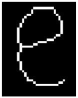
Structuring element W

44



34

only fit in the center!

open(f^c, W)=dilation(erosion(f^c),W), W)

Closing I

1. In general, closing also smoothes the object contour.

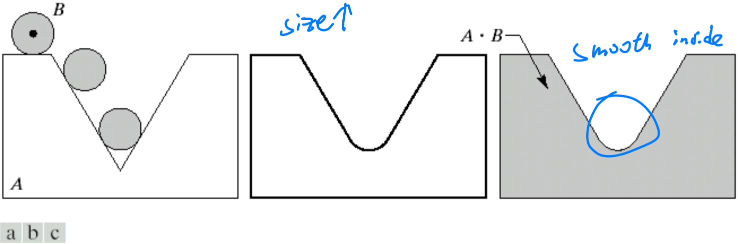


FIGURE 9.9 (a) Structuring element B "rolling" on the outer boundary of set A. (b) Heavy line is the outer boundary of the closing. (c) Complete closing (shaded).

2. Definition

$$A \bullet B = (A \oplus B) \ominus B.$$

Closing II

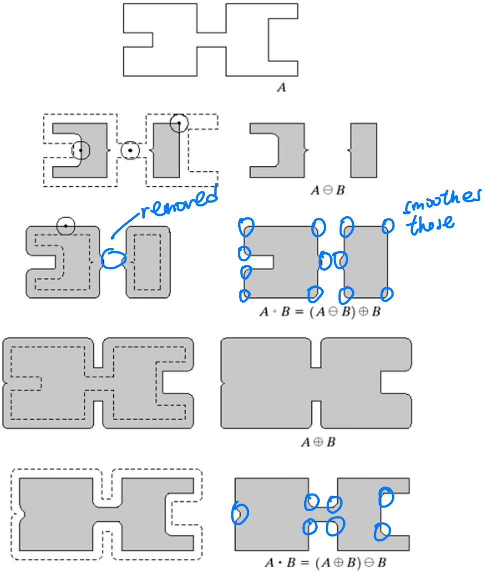
3. It represents dilation of A by B, followed by erosion of the result by B.
4. Imagine that you roll the ball outside the object instead of rolling the ball inside the object.
5. Subimage property: A is a subset (subimage) of $A \bullet B$.
6. Convergence property

$$(A \bullet B) \bullet B = A \bullet B.$$

7. Closing tends to enlarge the boundaries of the forground regions, and shrinks background color holes within this region, but less destructive than dilation of the original boundary shape.

a	
b	c
d	e
f	g
h	i

FIGURE 9.10 Morphological opening and closing. The structuring element is the small circle shown in various positions in (b). The dark dot is the center of the structuring element.



Comparison between opening and closing

The operation keeps the general shape but smooths w.r.t.

- opening $\to$ object
- closing $\to$ background

Opening and closing are dual

$$(A \bullet B)^c = A^c \circ \hat{B}.$$

= closing with background

Example of closing and opening

remove black!

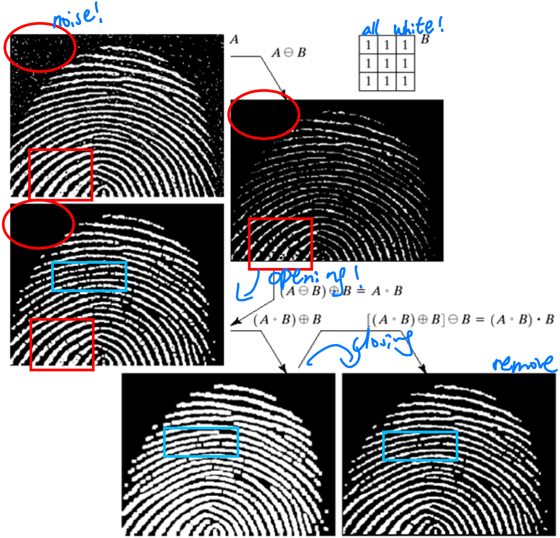


FIGURE 9.11
(a) Noisy image. (c) Eroded image. (d) Opening of A. (d) Dilation of the opening. (e) Closing of the opening. (Original image for this example courtesy of the National Institute of Standards and Technology.)

白色

same size
remove all the noise